

Prop-fn-compleja-derivable

Proposición 1 (Propiedades de la $\mathbb{C}$ -derivabilidad).

Sean $f, g: \mathbb{C} \longrightarrow \mathbb{C}$ dos funciones, entonces

- (1) f $\mathbb{C}$ -derivable en $z_0 \in \mathbb{C} \implies f$ es continua en z_0 .
- (2) f, g $\mathbb{C}$ -derivables en $z_0 \in \mathbb{C} \implies f \pm g, f \cdot g, f/g$ ($g(z_0) \neq 0$) son $\mathbb{C}$ -derivables en z_0 :
 - (a) $(f \pm g)'(z_0) = f'(z_0) \pm g'(z_0)$.
 - (b) $(f \cdot g)'(z_0) = f'(z_0)g(z_0) + f(z_0)g'(z_0)$.
 - (c) $(f/g)'(z_0) = \frac{f'(z_0)g(z_0) - f(z_0)g'(z_0)}{g^2(z_0)}$.
- (3) Regla de la cadena: f $\mathbb{C}$ -derivable en $z_0 \in \mathbb{C}$, g $\mathbb{C}$ -derivable en $f(z_0) \implies g \circ f$ es $\mathbb{C}$ -derivable en z_0 y $(g \circ f)'(z_0) = g'(f(z_0)) f'(z_0)$.

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